Architectural Tradeoffs in Low-Latency Algorithmic Trading with Predictive Signals

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March 5, 2026

1 Introduction

Financial markets have increasingly become dominated by automated trading systems that make decisions and execute trades at extremely high speeds. These systems rely on algorithmic strategies that analyze market data and react within milliseconds or even microseconds. As a result, the design of trading infrastructure has become just as important as the predictive models used to generate trading signals. Even when predictive models are accurate, delays in processing or order execution can eliminate any potential advantage in highly competitive markets.

Recent advances in machine learning have led to significant interest in applying predictive models to financial markets. Researchers have explored techniques ranging from classical statistical models to deep learning architectures for forecasting stock price movements. These models often attempt to identify patterns in historical price data, macroeconomic indicators, or market sentiment signals in order to generate short-term predictions about market behavior.

However, predictive accuracy alone does not guarantee profitability in financial markets. In practice, the ability to act on predictions depends on the latency of the trading system and the structure of the market itself. Modern electronic markets operate using limit order books and price-time priority mechanisms, meaning that orders arriving even slightly later than competitors may lose execution priority. This creates a critical interaction between prediction, system architecture, and market microstructure.

The goal of this project is to study these architectural tradeoffs by developing an event-driven trading simulation system. The system replays historical market data, processes events sequentially, generates short-term predictive signals, and measures the impact of latency and execution constraints on simulated trading outcomes. By focusing on the infrastructure that connects predictive models with market execution, this work aims to bridge the gap between machine learning research and practical trading system design.

In addition to predictive modeling challenges, the operational environment of financial markets introduces additional constraints. Modern exchanges operate at extremely high speeds and rely on complex technological infrastructures that process millions of messages per second. Trading systems must be designed to handle high-frequency streams of constant market data while maintaining a deterministic order and minimizing latency.

These constraints mean that the architecture of the trading system itself plays a crucial role in determining whether predictive signals can actually be translated into profitable trades. Even small delays in processing market data or submitting orders can result in lost queue position, reduced fill rates, and increased exposure to adverse price movements. Moreover, understanding the interaction between predictive models and trading system architecture has become an important area of study in quantitative finance and financial engineering.

2 Background

Understanding algorithmic trading systems requires familiarity with the structure of modern financial markets. Most electronic exchanges operate using a limit order book, where buy and sell orders are organized by price and time of submission. When traders submit limit orders, they specify a price at which they are willing to buy or sell an asset. These orders are placed into a queue and executed when matching orders arrive on the opposite side of the market.

The concept of price-time priority plays a central role in these systems. Orders with better prices are executed first, but when multiple orders exist at the same price level, they are executed in the order they were received. This means that execution speed can significantly affect trading outcomes. Even small delays in order submission can cause a trader to lose queue position, reducing the likelihood of execution. Harris provides a detailed overview of these market microstructure mechanisms and their implications for trading strategies [5].

HFT trading systems are specifically designed to minimize such delays. According to Aldridge, high-frequency trading firms routinely invest heavily in low-latency infrastructure, including using specialized hardware, software, and proximity hosting, trying to be as close to the exchange servers as possible [1]. These trading systems monitor and process incoming market data and update trading decisions in real time in microseconds. The performance of such systems depends not only on the accuracy of prediction models but also on the efficiency of event processing, memory access, and network communication.

Cartea et al. [8] further emphasizes that successful algorithmic trading strategies must always account for both predictive signaling (BUY/SELL/HOLD) and market impact. Trades can influence market prices, and poorly designed strategies may result from terrible selection, where informed traders exploit predictable behavior. As a result, realistic simulation environments must inlcude all aspects of market microstructure when evaluating a given trading algorithm.

3 Related Work

A large body of research has explored the use of machine learning techniques for financial market prediction. Early approaches relied primarily on linear models and statistical methods, but recent work has increasingly focused on deep learning architectures capable of capturing complex temporal patterns in financial data. Fischer and Krauss demonstrate that Long Short-Term Memory (LSTM) neural networks can outperform traditional machine learning models in predicting stock market returns using historical price data [3]. Their work

highlights the potential of deep learning models to capture long-term dependencies in financial time series.

Similarly, Gu, Kelly, and Xiu conduct a comprehensive empirical study of machine learning methods applied to asset pricing [4]. Their research evaluates a wide range of machine learning algorithms, including neural networks and tree-based models, using a large set of financial predictors. The authors find that machine learning approaches can improve predictive performance relative to traditional linear models, particularly when working with high-dimensional datasets.

More recent research has explored the integration of additional data sources into predictive models. Patsiarikas et al. [6] examine the use of macroeconomic indicators, technical market signals, and sentiment data for stock market forecasting. Their results suggest that combining multiple categories of data can improve predictive performance compared to using price data alone. However, these studies typically focus on prediction accuracy rather than the operational constraints involved in executing trading strategies in real markets.

Other research emphasizes the importance of market microstructure and execution dynamics. Bouchaud, Farmer, and Lillo analyze how markets absorb supply and demand changes over time, demonstrating that trading activity itself can influence price formation [2]. Their work highlights the importance of understanding how trades interact with market liquidity and order flow when designing trading strategies.

While the literature provides extensive research on predictive modeling and market behavior, fewer studies examine the interaction between predictive signals and the architecture of trading systems. Real-time stream processing research provides insight into the design of low-latency systems capable of processing high-frequency data streams. Stonebraker et al. [7] identify key requirements for real-time data processing systems, including low latency, deterministic processing, and efficient memory management. These principles are directly relevant to the design of algorithmic trading platforms.

This project builds on these ideas by combining predictive modeling concepts with a systems-oriented approach to trading simulation. Rather than focusing solely on prediction accuracy, the project investigates how architectural decisions influence the ability of trading systems to act on predictive signals under realistic market conditions.

4 Methodology

This project begins by constructing the foundational infrastructure required for an event-driven trading system simulation. At this stage, the focus is on market data ingestion and deterministic event replay, which form the basis for analyzing architectural tradeoffs in low-latency trading systems.

4.1 Market Data Acquisition

Historical market data is downloaded using the Yahoo Finance API through the `yfinance` Python library. The current implementation retrieves recent S&P 500 index data at one-minute intervals and stores timestamped price and volume information in CSV format. The

preprocessing step flattens multi-index columns, standardizes column names, and converts timestamps to integer Unix time for consistent processing.

This approach ensures reproducibility and establishes a structured input format for the simulation pipeline. Using historical replay rather than live feeds aligns with common practices in quantitative finance research, where controlled backtesting environments are used to evaluate trading logic [1, 8].

4.2 Event Abstraction

To support an event-driven architecture, market data is abstracted into discrete events. An `Event` data structure is defined with three fields: timestamp, event type, and payload. Event types are enumerated to allow future extension for order submission, cancellation, and execution events.

Each market update is represented as a structured event object, enabling uniform handling of data within the system. This abstraction follows principles of real-time stream processing systems, where events are processed sequentially and deterministically [7]. Event abstraction is critical for modeling realistic trading environments, where exchanges operate as event-driven systems governed by limit order book mechanics [5].

4.3 Deterministic Market Replay

A replay engine iterates sequentially over historical market data and converts each row into a `MARKET_UPDATE` event. The replay mechanism maintains an internal index and exposes methods to check for remaining events and retrieve the next event in chronological order.

Deterministic replay prevents look-ahead bias and ensures that events are processed strictly in historical sequence. This is essential when studying the interaction between prediction and execution, since improper ordering can artificially inflate model performance [2]. Although a full event dispatcher and execution engine have not yet been implemented, the replay system establishes the primary input stream for the future low-latency architecture.

5 Conclusion

References

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